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MRI-Based Quantification of Intratumoral Heterogeneity for Predicting Progression-Free Survival in Patients with Lung Cancer Brain Metastasis Receiving Radiotherapy

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ABSTRACT

BACKGROUND AND PURPOSE: Our aim was to investigate the potential of using MRI-based habitat features for predicting progression-free survival (PFS) in patients with lung cancer brain metastasis (LCBM) receiving radiotherapy.

MATERIALS AND METHODS: One hundred and forty-six lesions from 68 patients with LCBM receiving radiotherapy were retrospectively reviewed and divided into training, random test (R-test), and time-independent test (TI-test) cohorts. Conventional radiomics and habitat features were extracted from the whole-tumor area and tumor subregions, respectively. Different machine learning risk models for predicting PFS were developed on the basis of clinical, radiomics, and habitat features, and their combination (clinical + habitat), respectively. The performance of the risk models was evaluated using the concordance index (C-Index) and Brier scores. The Kaplan-Meier curve was used to assess the prognostic value of the models.

RESULTS: The habitat risk model achieved the best prediction ability among 4 different risk models in the TI-test cohort (C-Index: 0.716; 95% CI, 0.548–0.890). Additionally, the habitat and radiomics risk models outperformed the clinical risk model in the training (C-Index: 0.721–0.762 versus 0.697) and TI-test cohorts (C-Index: 0.630–0.716 versus 0.377).

CONCLUSIONS: A habitat risk model based on intratumoral heterogeneity could be a reliable biomarker for predicting PFS in patients with LCBM receiving radiotherapy.

ABBREVIATIONS: BM = brain metastasis; C-Index = concordance index; Cox PH = Cox proportional hazard; FAE = FeAture Explorer; ICC = intraclass correlation coefficient; LCBM = lung cancer brain metastasis; nnFAE = nnFeAture Explorer; PFS = progression-free survival; R-test = random test; TICE = postcontrast TI-weighted imaging; TI-test = time-independent test

Metastasis is the leading cause of death in patients with lung cancer, and the brain is one of the most common sites for lung cancer metastasis.¹ Radiotherapy is one of the most efficient treatment strategies for lung cancer brain metastasis (LCBM).² Although standard treatment strategies are used, approximately 30% of patients experience local failure after treatment.³ Therefore, accurate pretherapeutic prediction of

the treatment outcome of the patient with LCBM receiving radiotherapy is of great clinical value.

Intratumoral heterogeneity, which generally exists in brain metastasis (BM), has been recognized as the key contributor to intracranial progression and radiation resistance. Due to this intratumoral heterogeneity, tumors with variable microenvironments may have discrepant responses to standard treatment strategies across different tumor lesions and even different tumor subregions.^{4–6} Thus, precise assessment of intratumoral heterogeneity may have promising potential in predicting tumor treatment response and improving therapeutic decision-making.

Factors such as the Karnofsky Performance Status, patient age, tumor volume, the presence of extracranial metastases, and the number of BMs are used and combined in the literature to determine prognostic indices that have been proposed for the prediction of survival in patients with BM.⁷ Several new MRI-based techniques, such as chemical exchange saturation transfer, PWI, and MRS, have shown promise for predicting tumor response or prognosis.^{8–10} However, they did not reveal the potential role of intratumoral heterogeneity in the treatment response prediction of BM.

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SUMMARY

PREVIOUS LITERATURE: Several new MRI-based techniques, such as chemical exchange saturation transfer, PWI, and MRS, have shown promise for predicting tumor response or prognosis. Also, radiomics features derived from MRI of LCBM have been used to predict the treatment response to radiotherapy by measuring the characteristics of the whole-tumor region in previous studies. However, they did not reveal the potential role of intratumoral heterogeneity on treatment outcome prediction of brain metastasis.

KEY FINDINGS: A risk model based on habitat feature–quantified intratumoral heterogeneity could be a reliable biomarker to predict PFS in patients with LCBM receiving radiotherapy.

KNOWLEDGE ADVANCEMENT: The intratumoral heterogeneity of LCBM could be quantified via MRI-based habitat features. A habitat risk model enabled satisfactory PFS prediction for patients with LCBM receiving radiotherapy, having considerable potential for supporting decision-making in clinical practice.

Radiomics features derived from MRI of LCBM have been used to predict the treatment response to radiotherapy by measuring the characteristics of the whole-tumor region.^{11,12} This method regards the tumor as a whole and neglects regional heterogeneity within the tumor. Furthermore, prior approaches calculate the features per sequence without combining multiple sequences (which reflect variable information) simultaneously or quantifying the spatial relationships among voxels. Consequently, imaging biomarkers that reflect the overall characteristics of the whole-tumor region may be inadequate for fully revealing intratumoral heterogeneity and precisely predicting tumor treatment response.

Habitat analysis, an emerging approach, explicitly divides tumors into subregions containing clusters of voxels with similar characteristics, allowing better quantification of tumor heterogeneity. With the assistance of unsupervised algorithms and multiparameter MRI, habitat features can depict intratumoral heterogeneity in a more interpretable and multidimensional way.^{13–15} Recently, several studies have reported that intratumoral heterogeneity quantified with habitat features is strongly related to treatment resistance in glioblastoma, oropharyngeal cancer, and breast cancer.^{16–18} However, the potential of identifying LCBM at risk of local failure to improve patient management in LCBM has not been validated.

Therefore, the purpose of this study was to develop and validate a predictive model on the basis of MRI-based habitat features that could quantify intratumoral heterogeneity to predict progression-free survival (PFS) in patients with LCBM before radiotherapy.

MATERIALS AND METHODS

Patient Data

This study was approved by the institutional review board of the First Affiliated Hospital of Fujian Medical University (MRCTA, ECFAH of FMU [2020] 192), and informed consent was waived due to its retrospective nature. On the basis of the inclusion and exclusion criteria shown in Fig 1A, we retrospectively enrolled patients with LCBM who received radiotherapy between January 2018 and November 2022. When patients underwent radiotherapy for multiple lesions, up to 3 lesions were considered for inclusion, prioritized by

size.¹⁴ The patients who received radiotherapy from January 2018 to February 2022 were included as the primary cohort and were randomly assigned to training and random test (R-test) cohorts at a ratio of 7:3 by reference to the number of progression/nonprogression cases. Patients who received radiotherapy from March 2022 to November 2022 were defined as the time-independent test (TI-test) cohort.

Baseline clinical characteristics, including age, sex, the histologic subtype of lung cancer, number of metastases, targeted drug treatment, history of smoking, neurologic symptoms, Karnofsky Performance Status score, tumor location, extracranial metastasis, radiotherapy technique, and the biologically effective dose, were collected from our medical records.

Treatment Response Assessment and Follow-Up

The baseline imaging was acquired before the initiation of radiotherapy. Subsequently, MRI follow-up scans were usually performed at intervals of every 2–3 months. Tumor treatment response was assessed on the basis of the Response Assessment in Neuro-Oncology Brain Metastases (RANO-BM) criteria at the lesion level according to previous literature.¹⁹ A maximum unidimensional diameter of the tumor exceeding 20% on follow-up postcontrast T1-weighted imaging (T1CE) was defined as progression. The response assessment was conducted by 2 radiologists with 5 and 20 years of experience in BM diagnosis. Disagreements were reviewed by the 2 radiologists, and a consensus was reached by discussion.

PFS was selected as the clinical end point in this study, which was defined as the time from the date of receiving radiotherapy to the date of progression or the last imaging examination during follow-up. The cutoff follow-up time point of this study was January 23, 2024.

MRI Data Acquisition and Preprocessing

For each patient, pretreatment T1CE and FLAIR images were obtained on 1.5T or 3T MR scanners, and the details of the acquisition parameters for MRI are shown in the Supplemental Data.

All MR images underwent a standard image-preprocessing protocol, including skull-stripping, N4 bias field correction, registration, and intensity normalization. The first step is to perform skull-stripping using FMRIB Software Library, Version 5.0.8,

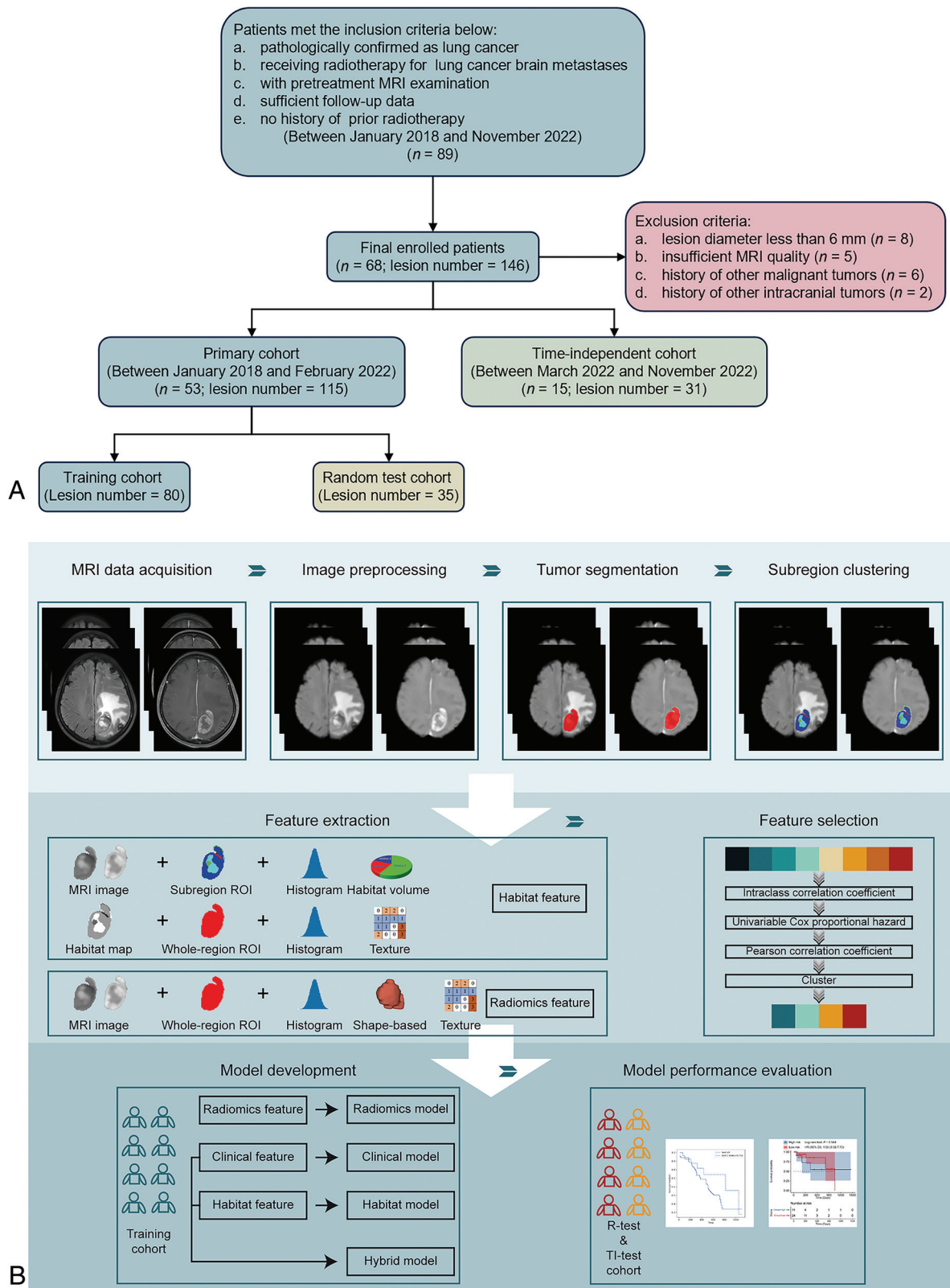


FIG 1. Patient enrollment (A) and study analysis (B) flowchart. ROI indicates region of interest; R-test, random test; TI-test, time-independent test.

software (FSL; <http://www.fmrib.ox.ac.uk/fsl>). N4 bias field correction was applied to all images before further processing to reduce intensity inhomogeneity using SimpleITK (Version 2.3.1; <https://simpleitk.org/>). Additionally, the Advanced Normalization

Tools (Version 2.5.0; <https://github.com/ANTsX/ANTs>) were used to align FLAIR sequences to the T1CE sequences. Ultimately, the pixel intensities of the resulting gray-scale images were normalized to the range of 0–255 with SimpleITK.

Table 1: Various risk models and selected features in each model

Models	Variables	Hazard Ratio (95% CI)	P Value
Clinical	Sex	1.62 (0.64–4.15)	.31
	Targeted_drug_treatment	0.47 (0.19–1.21)	.12
Radiomics	T1CE_original_gldm_DependenceEntropy	0.82 (0.06–11.29)	.88
	T1CE_original_glrIm_GrayLevelNonUniformityNormalized	1.40 (0.07–29.64)	.83
	T1CE_original_glcM_Id	1.18 (0.10–14.39)	.90
	T1CE_original_glcM_InverseVariance	0.74 (0.07–8.48)	.81
	T1CE_original_glcM_SumEntropy	0.77 (0.04–13.37)	.86
	T1CE_original_glszm_SizeZoneNonUniformityNormalized	4.78 (0.51–45.16)	.17
	FLAIR_original_glszm_LargeAreaHighGrayLevelEmphasis	9.73 (1.38–68.63)	.02
Habitat	FLAIR_habitat2_Energy	1.93 (0.19–19.83)	.58
	T1CE_habitat3_Variance	0.35 (0.06–2.14)	.26
	T1CE_habitat5_Energy	2.84 (0.33–24.73)	.34
	habitat2_Percent	3.52 (0.75–16.57)	.11
	habitatmap_original_glszm_ZoneVariance	2.84 (0.31–25.74)	.35
Hybrid	Targeted_drug_treatment	0.59 (0.22–1.59)	.30
	Age	2.27 (0.36–14.14)	.38
	Histologic_subtype_3	1.53 (0.41–5.71)	.53
	T1CE_habitat3_Entropy	0.62 (0.14–2.78)	.53
	T1CE_habitat5_Energy	1.67 (0.17–16.64)	.66
	habitat2_Percent	3.05 (0.70–13.18)	.14
	habitat3_Percent	0.68 (0.14–3.40)	.64
	habitatmap_original_glszm_ZoneVariance	4.91 (0.59–40.69)	.14

Tumor Segmentation

ROIs of lesions were segmented using 3D Slicer software (Version 5.0.3; <http://www.slicer.org>). These tumor ROIs were manually annotated layer by layer along the tumor margins by a radiologist (radiologist A) with 5 years of diagnostic experience in BM MRI. The ROIs were delineated according to an enhanced margin of tumor on T1CE images while excluding T2-FLAIR perienhancing alterations. Both cystic and necrotic areas of the tumors were included in the ROIs. Afterward, all MRIs and ROIs were resampled to $1 \times 1 \times 1$ mm with SimpleITK.

Thirty lesions were randomly chosen for repetitive tumor ROI segmentation, performed by 2 radiologists (radiologists A and B, with 5 and 7 years of experience in BM MRI, respectively) to explore the intraobserver and interobserver stability of features.

Subregion Clustering and Feature Extraction

A total of 198 radiomics features were extracted from the whole-tumor region, including 17 histogram-based and 75 texture-based features for each sequence and 14 shape-based features of the whole-tumor region.

Then we analyzed all voxels of T1CE and FLAIR using the nonsupervised clustering method, K-means, to split the ROIs into 2, 3, 4, and 5 cluster regions by the in-house-developed software nnFAE (Version 0.1.0) based on Python (Version 3.9.10) and scikit-learn (Version 1.6.0; <https://github.com/scikit-learn/scikit-learn>). The clustering results were evaluated using the average Calinski–Harabasz score to determine the optimal number of clusters ($\kappa = 5$ in this study, Supplemental Data).

After subregion clustering, habitat features were extracted from the intratumoral subregions in this study. Initially, 36 features were extracted from each subregion of the tumor regions, including 34 histogram features (17 features for each sequence), the volume of voxels, and the volume proportion of each subregion. To further measure tumor heterogeneity, we constructed a habitat map in which the intensity values of voxels were replaced

with the previously identified habitat index. We adopted the concept of quantitative imaging features from radiomics studies to measure the spatial heterogeneity among the identified subregions.²⁰ Therefore, 44 features were calculated from the habitat map, including 4 histogram features, 24 gray-level co-occurrence matrix features, and 16 gray-level size zone matrix features.

The habitat and radiomics features extracted in this study complied with the image biomarker standardization initiative. The details of the habitat and radiomics features extracted in this study are summarized in the Supplemental Data.

Feature Selection and Modeling

Initially, only radiomics and habitat features with both intra- and interobserver intraclass correlation coefficients (ICCs) > 0.75 were retained for further analysis. Then, univariable Cox proportional hazard (Cox PH) analysis was conducted to identify potential risk factors from the clinical, radiomics, and habitat features, and factors with P values $< .10$ were retained for subsequent survival analysis using FeAture Explorer (FAE, Version 0.6.0).²¹ Then the features were standardized with the min-max method due to the significant dimension difference. To reduce the dimensions of the row space of the feature matrix, we used the Pearson correlation coefficient for each pair of features. If the Pearson correlation coefficient exceeded 0.99, we randomly removed 1 of the correlated features. Then, these features were input into a clustering procedure to select the most relevant feature subset. To prevent overfitting of the model, we limited the maximum number of the selected features in the subset to no more than 10% of the sample size used for training in this study. The selected feature subsets were subsequently trained using the Cox PH model with 5-fold cross-validation to determine the optimal number of features. Ultimately, the clinical-, radiomics-, and habitat-based machine learning risk models were built to predict the PFS of lesions.

Table 2: Performance of different risk-prediction models

Models	Training Cohort		R-Test Cohort		TI-Test Cohort	
	C-Index (95% CI)	Brier Score (95% CI)	C-Index (95% CI)	Brier Score (95% CI)	C-Index (95% CI)	Brier Score (95% CI)
Clinical	0.697 (0.587–0.817)	0.138 (0.105–0.164)	0.719 (0.484–0.889)	0.177 (0.110–0.225)	0.377 (0.266–0.519)	0.206 (0.136–0.296)
Radiomics	0.762 (0.612–0.914)	0.145 (0.111–0.178)	0.700 (0.545–0.829)	0.276 (0.113–0.440)	0.630 (0.309–0.900)	0.186 (0.088–0.313)
Habitat	0.721 (0.598–0.888)	0.133 (0.104–0.159)	0.716 (0.543–0.877)	0.273 (0.104–0.514)	0.716 (0.548–0.890)	0.171 (0.086–0.292)
Hybrid	0.767 (0.638–0.898)	0.105 (0.067–0.134)	0.748 (0.575–0.910)	0.268 (0.082–0.536)	0.590 (0.417–0.785)	0.228 (0.124–0.371)

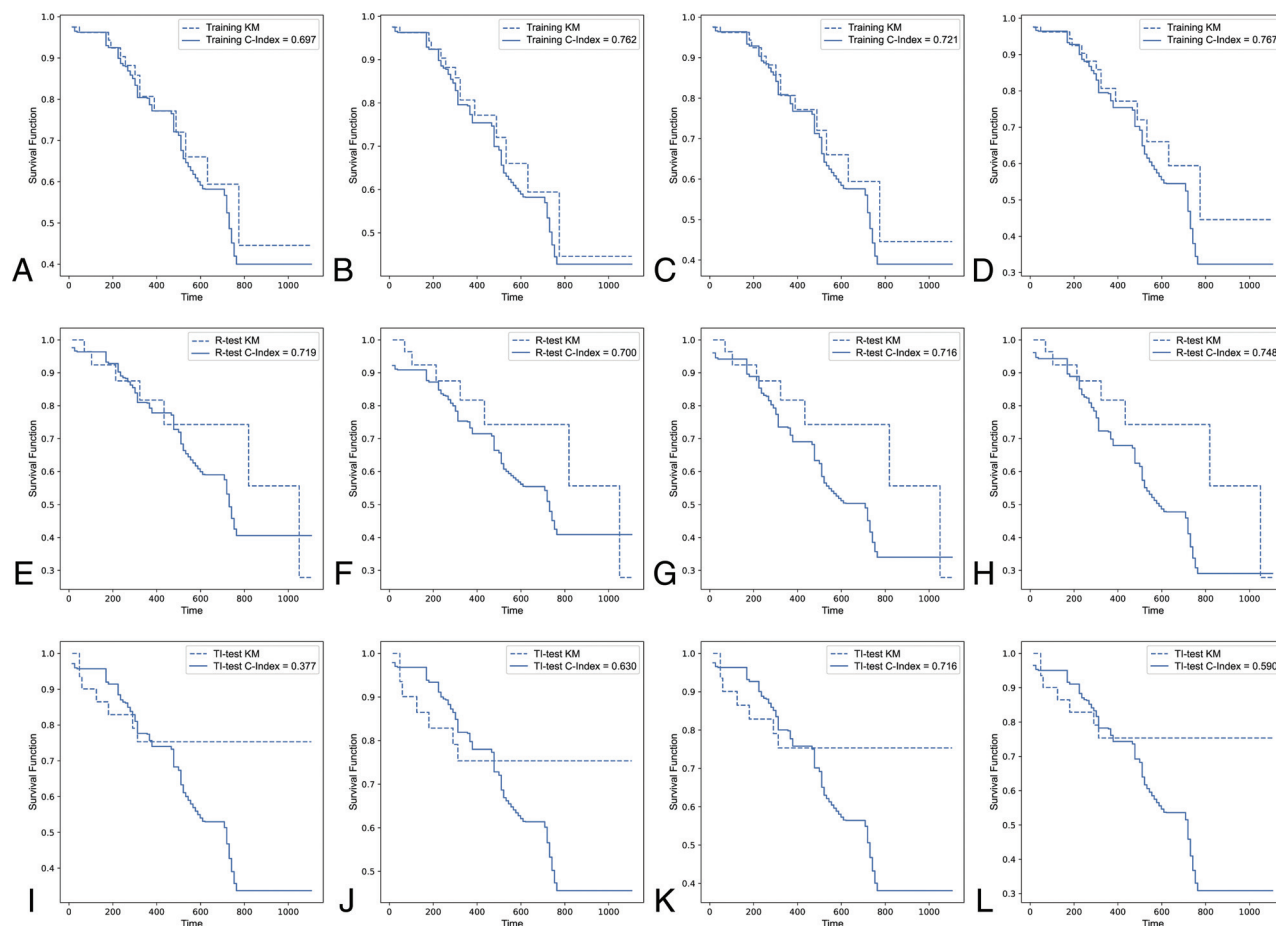


FIG 2. The actual survival curve and the model-predicted survival curve in the training (first row), R-test (second row), and TI-test (third row) cohorts, defined by the clinical (first column), radiomics (second column), habitat (third column), and hybrid (fourth column) models. KM indicates Kaplan–Meier.

For hybrid risk model construction, clinical and habitat features with P values under .10 in the univariate Cox PH analysis were integrated. The hybrid model was developed following a pipeline consistent with that used for the other risk models, including feature matrix normalization (min-max method), Pearson correlation coefficient for featured dimension reduction, cluster for feature selection, and Cox PH for model fit with 5-fold cross-validation.

Fig 1B shows the workflow of this study.

Prognostic Value of the Risk Models

For each of the 4 risk models, its potential association with PFS was evaluated in the training, R-test, and TI-test cohorts using Kaplan-Meier analysis. The lesions in the training cohort were

stratified into high-risk and low-risk groups based on the optimal cutoff of the prediction probability of models with X-tile software (Version 3.6.1; <https://x-tile.software.informer.com/3.6/>). The cutoff values were then applied to the R-test and TI-test cohorts. A log-rank test was used to assess the differences in survival between the 2 risk groups.

Statistical Analysis

Statistical analysis was performed using Python (Version 3.11) and R software (Version 4.4.2; <http://www.r-project.org/>). Categorical variables were compared using the χ^2 or Fisher exact test. Continuous variables were compared using the Student t test or the Mann–Whitney U test as appropriate. The clinical features among groups were compared with the CBCgrps (Version 2.8.2) package.

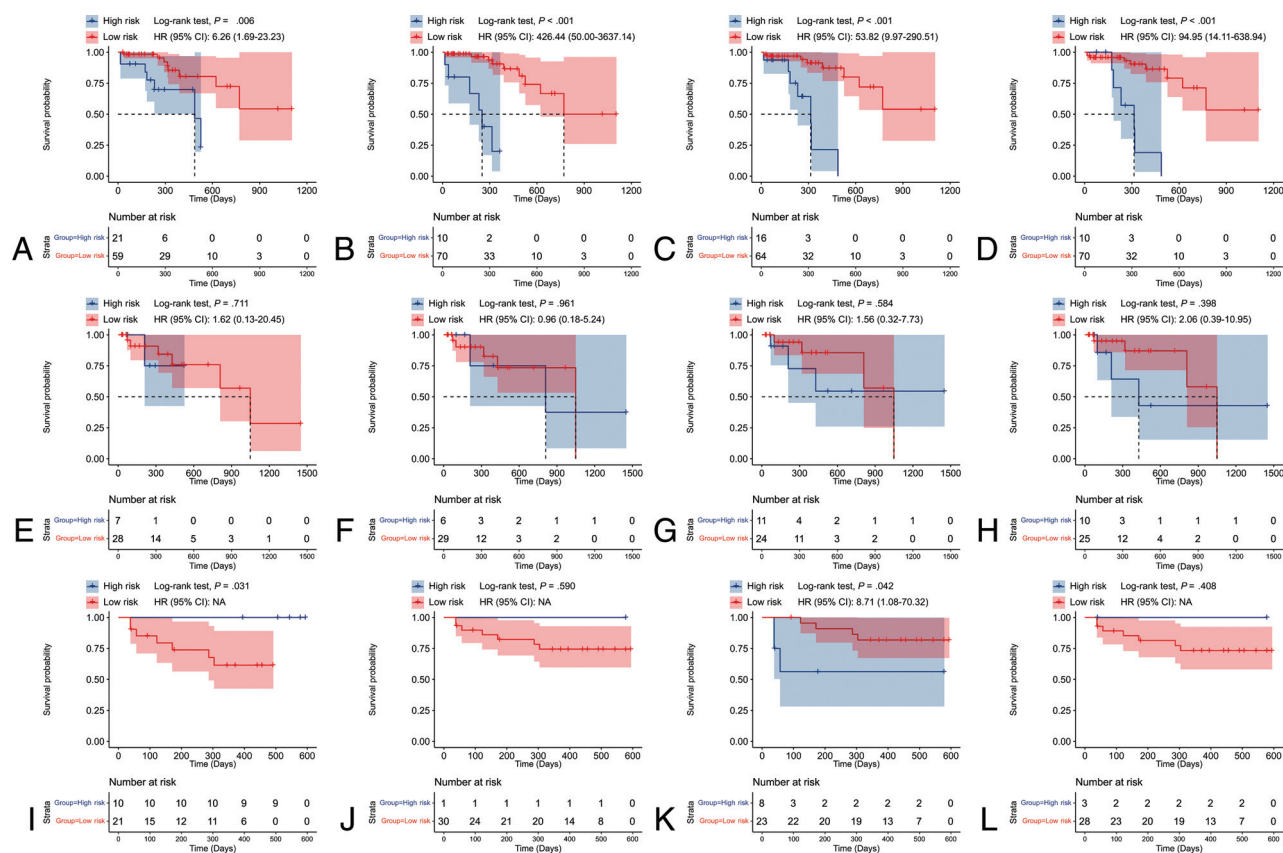


FIG 3. Kaplan-Meier survival analysis of the survival outcomes of different risk groups in the training (first row), random test (second row), and time-independent test (third row) cohorts, defined by the clinical (first column), radiomics (second column), habitat (third column), and hybrid (fourth column) models. HR indicates hazard ratio; NA indicates not applicable.

The Calinski-Harabasz score was calculated with the scikit-learn (Version 1.4.2) package. The ICC analysis of features was implemented with the Pingouin (Version 0.5.3; <https://pingouin-stats.org/build/html/index.html>) package. The univariate Cox PH analysis and Kaplan-Meier analysis were conducted with survival (Version 3.7-0) and survminer (Version 0.5.0) packages. The features normalization, Pearson correlation coefficient analysis, cluster analysis for feature dimension reduction, and Cox PH model development were constructed using the survival analysis module of FAE. The predictive performance of the models was evaluated using the concordance index (C-Index) and Brier score by FAE automatically.

RESULTS

Patient Characteristics

A total of 146 lesions from 68 patients with LCBM were retrospectively identified. No significant difference was observed in clinical characteristics between the training and R-test cohorts (Supplemental Data); no significant difference was found among the 3 cohorts, except for sex ($P = .003$) and biologically effective dose (BED) ($P = .002$) (Supplemental Data).

Feature Selection and Development of Prediction Models

After the ICC analysis, 192 radiomics features and 187 habitat features were retained for subsequent analysis. Univariate Cox PH analysis identified 4 clinical, 26 radiomics, and 11 habitat

features potentially associated with lesions' PFS ($P < .10$), as detailed in the Supplemental Data. After dimensionality reduction and feature selection, the clinical, radiomics, habitat, and hybrid risk models incorporated 2, 7, 5, and 8 features, respectively. The selected features for each risk model are summarized in Table 1.

Performance Evaluation of the Risk Models

The models' performance is shown in Table 2 and Fig 2. The habitat risk model demonstrated superior discrimination compared with the other risk models in the TI-test cohort (C-Index: 0.716, 95% CI: 0.548–0.890; Brier score: 0.171, 95% CI: 0.086–0.292). The hybrid model outperformed the other models in the R-test cohort (C-Index: 0.748, 95% CI: 0.575–0.910; Brier score: 0.268, 95% CI: 0.082–0.536). Additionally, the habitat and radiomics risk models showed better performance than the clinical risk model in the TI-test cohort (C-Index: 0.630–0.716 versus 0.377).

Prognostic Value of the Risk Models

In the TI-test cohort, only the PFS in the risk groups defined by the habitat risk model showed significant differences ($P = .042$) (Fig 3).

Figure 4 presents the habitat characteristics of tumors in 2 cases.

DISCUSSION

In this study, we developed and validated a risk model based on habitat feature-quantified intratumoral heterogeneity to predict

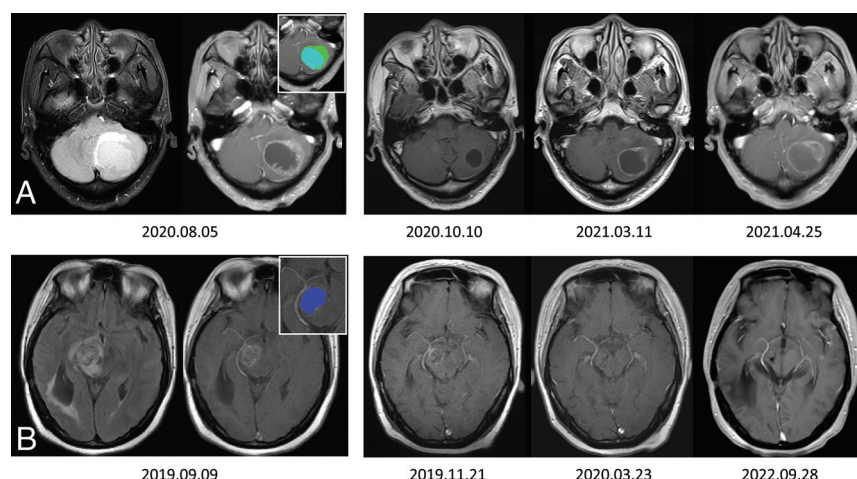


FIG 4. Example of tumor habitat in 2 cases. Case 1 (A) of the progression group (baseline MRI: 2020.08.05, and follow-up MRI: 2020.10.10, 2021.03.11, 2021.04.25; PFS 210 days). Case 2 (B) of the nonprogression group (baseline MRI: 2019.09.09, and follow-up MRI: 2019.11.21, 2020.03.23, 2022.09.28; PFS 1105 days).

PFS in patients with LCBM receiving radiotherapy. The habitat risk model obtained satisfactory results (C-Index: 0.716), outperforming the clinical, radiomics, and hybrid risk models (C-Index: 0.377–0.630) in the TI-test cohort.

Quantitative MRI biomarkers extracted from traditional or advanced MRI that characterize the cellular, biophysical, microstructural, and metabolic features of tumors have been proposed as promising markers for the prediction of treatment outcomes in those with BMs.^{9,22} However, they did not reveal the potential role of intratumoral heterogeneity on BM's treatment response prediction. Lee et al²³ discovered that the structural and physiologic MRI habitat volume was linked to BM recurrence risk and site after stereotactic radiosurgery. Our study also showed similar results and yielded a comparable prediction ability with the previous study (highest C-Index, previous study versus our study, 0.71 versus 0.716).

The habitat features extracted in the present study were derived by combining T1CE and FLAIR images. In previous radiomics studies,^{24,25} the radiomics features derived from T1CE and FLAIR images were calculated separately and combined only at the level of the overall tumor area. In our study, each voxel in the tumor area was characterized by T1CE and FLAIR simultaneously. We combined multiple physiologic properties at the level of a single voxel. Consequently, with this framework, both the physiologic heterogeneity and the spatial heterogeneity of the LCBM could be fully identified.

Compared with the prognostic predictive ability of the hybrid risk model, the habitat risk model showed better predictive performance in the TI-test cohort, indicating that the addition of clinical features failed to improve the prediction efficiency of the habitat risk model in predicting PFS in patients with LCBM receiving radiotherapy, which was inconsistent with the findings of previous studies.^{18,26,27} One explanation could be that the prediction ability of the clinical features is not as stable as that of habitat features making the relatively low C-Index of clinical modes in the TI-test data set, which in turn weakens the predictive ability of the hybrid risk model.

This study has several limitations. First, the study was retrospective and single-center in nature and involved relatively small sample sizes. However, multiple test cohorts and multiple statistical metrics were used to evaluate the model performance, which improved the reliability of our results. Another limitation of our study was the determination of lesion end points. Due to the low resection rate of BM in this cohort upon progression, local failure was determined radiographically based on a measure of tumor size according to the RANO-BM criteria. Histopathologic analysis is the gold standard for determining radiation necrosis versus tumor progression. Third, as a retrospective study, the slice thickness and gap varied among different scanners. Although the

standard resample preprocessing was applied, a thinner slice thickness (eg, 1 mm) may improve the accuracy of assessment of tumor characteristics. In the future, a prospective study with thinner slice thickness may be adapted to improve the model performance. Lastly, the use of cluster data may introduce some problems such as intracluster correlation, reduced statistical power, and potential bias due to confounding factors. The approach of our study is consistent with recent landmark studies, which also include multiple lesions from 1 patient as independent samples without additional statistical analyses.^{11,12,28} Future multicenter studies with larger cohorts will be conducted and a single-lesion-per-patient selection protocol will be employed to maintain sample independence while improving demographic representation.

CONCLUSIONS

In conclusion, the intratumoral heterogeneity of LCBM could be quantified via MRI-based habitat features. A habitat risk model enabled satisfactory PFS prediction for patients with LCBM receiving radiotherapy, holding considerable potential for supporting decision-making in clinical practice.

Disclosure forms provided by the authors are available with the full text and PDF of this article at www.ajnr.org.

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